

```

library(reshape2)

data <- ChickWeight

cat("First 10 records:\n")

print(head(data, 10))

# Sort by weight

sorted_data <- data[order(data$weight), ]

cat("\nSorted Data:\n")

print(head(sorted_data, 10))

# Melt the data

melted_data <- melt(

  data,

  id.vars = c("weight", "Time", "Chick", "Diet")

)

cat("\nMelted Data:\n")

print(head(melted_data, 10))

# Cast according to Diet

diet_average <- dcast(

  data,

  Diet ~ Time,

  value.var = "weight",

  fun.aggregate = mean

)

cat("\nAverage Weight by Diet and Time:\n")

print(diet_average)

```

RStudio interface showing R code execution and output.

Source Editor:

```
4 print(head(data, 10))
5 # Sort by weight
6 sorted_data <- data[order(data$weight), ]
7 cat("\nSorted Data:\n")
8 print(head(sorted_data, 10))
9 # Melt the data
10 melted_data <- melt(
11   data,
12   id.vars = c("weight", "Time", "Chick", "Diet")
13 )
14 cat("\nMelted Data:\n")
15 print(head(melted_data, 10))
16 # Cast according to Diet
17 diet_average <- dcast(
18   data,
19   Diet ~ Time,
20   value.var = "weight",
21   fun.aggregate = mean
22 )
23 cat("\nAverage Weight by Diet and Time:\n")
24 print(diet_average)
25
```

Environment:

Object	Variables
data	578 obs. of 4 variables
diet_average	4 obs. of 13 variables
melted_data	578 obs. of 4 variables
sorted_data	578 obs. of 4 variables

Console:

```
10 171 18 1 1

Sorted Data:
weight Time Chick Diet
196 35 2 18 1
26 39 2 3 1
195 39 0 18 1
293 39 0 27 2
305 39 0 28 2
317 39 0 29 2
365 39 0 33 3
401 39 0 36 3
543 39 0 48 4
13 40 0 2 1

Melted Data:
weight Time Chick Diet
1 42 0 1 1
2 51 2 1 1
3 59 4 1 1
4 64 6 1 1
5 76 8 1 1
6 93 10 1 1
7 106 12 1 1
8 125 14 1 1
9 149 16 1 1
10 171 18 1 1

Average Weight by Diet and Time:
Diet 0 2 4 6 8 10
1 1 41.4 47.25 56.47368 66.78947 79.68421 93.05263
2 2 40.7 49.40 59.80000 75.40000 91.70000 108.50000
3 3 40.8 50.40 62.20000 77.90000 98.40000 117.10000
4 4 41.0 51.80 64.50000 83.90000 105.60000 126.00000
12 14 16 18 20 21
1 108.5263 123.3889 144.6471 158.9412 170.4118 177.7500
2 131.3000 141.9000 164.7000 187.7000 205.6000 214.7000
3 144.4000 164.5000 197.4000 233.1000 258.9000 270.3000
4 151.4000 161.8000 182.0000 202.9000 233.8889 238.5556
>
```